Scientitif Computing with Python
Lesson 3: Scipy - Newtons Optimization

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Question: Using **Newton's** optimization method, minimize the function:

$$f(x, y) = x^2 + x^2y + y^2$$

Steps:

1. Compute the partial derivative of $f(x, y)$ with respect to x.

$$\frac{\partial f}{\partial x} = 2x + 2xy$$

2. Compute the partial derivative of $f(x, y)$ with respect to y.

$$\frac{\partial f}{\partial y} = x^2 + 2y$$

3. Gradient of the function $f(x, y)$, at point $(2, 1)$.

$$\nabla f(x, y) = (2x + 2xy, x^2 + 2y)$$

$$\nabla f(x, y) = (8, 6)$$

4. Compute $\frac{\partial^2 f}{\partial x^2}$, f_{xx}

$$\implies \frac{\partial f}{\partial x}(2x + 2xy) = 2 + 2y$$

5. Compute second derivative of $f(x, y)$, f_{yx}

$$\frac{\partial^2 f}{\partial y \partial x} = 2x$$

6. Compute the second partial derivative of $f(x, y)$, f_{xy}

$$f''_{xy} = \frac{\partial^2 f}{\partial x \partial y} = 2x$$

7. Compute second derivative of $f(x, y)$, f_{yy}.

$$\implies \frac{\partial^2 f}{\partial y^2} = 2$$

8. Now, the Hessian matrix, $\mathbf{H}$

$$\mathbf{H} = \begin{pmatrix} 2+2y & 2x \\ 2x & 2 \end{pmatrix}$$

$$\mathbf{H}(2,1) = \begin{pmatrix} 4 & 4 \\ 4 & 2 \end{pmatrix}$$

9. Compute $\mathbf{H}^{-1}$

$$\implies \frac{1}{-8}\begin{pmatrix} 2 & -4 \\ -4 & 4 \end{pmatrix} = \begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\mathbf{H}^{-1} = \begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

10. Prove that $\mathbf{H}^{-1}\mathbf{H} = I$

$$\implies \begin{pmatrix} 4 & 4 \\ 4 & 2 \end{pmatrix}\begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = I$$

$$= -\frac{1}{4}\begin{pmatrix} 4 \\ 4 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\implies \frac{1}{2}\begin{pmatrix} 4 \\ 4 \end{pmatrix} + -\frac{1}{2}\begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\therefore \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\implies \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \blacksquare$$

11. Solve for Newton's optimization, using the formula: $x_{new} = x_{old} - \dfrac{f'(x_{old})}{f''(x_{old})}$

$$\nabla f(2,1) = (8,6)$$

$$\mathbf{H}^{-1} = \begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\implies \mathbf{H}^{-1}\nabla f(2,1) = \begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} 8 \\ 6 \end{pmatrix} = \begin{pmatrix} -2+3 \\ 4-3 \end{pmatrix}$$

$$\therefore \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$